

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10: 1.3 THE PERIODIC TABLE

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have finished all 10 lessons on the Periodic Table. Now write your own complete explanation of the driving question: How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form? Use your Element Cards, your notes from each lesson, the Driving Question Board (DQB) and your Class Model (the Electron Arrangement Model) from Lesson 1 to Lesson 10. Write in complete sentences, use scientific words correctly (atomic number, outer electrons, group, period, noble gas, cation, anion, valency, oxidation number, radical), and give evidence from the activities to support every claim. Show your reasoning clearly: say what you saw, then explain why. Write your answers in the space under each prompt. Take your time and show what you have learned.

Part 1: How Is the Periodic Table Ordered?

In Lesson 1 you sorted the first 20 Element Cards by mass and by properties. (a) Explain how Mendeleev (1869) and Moseley (1913) each ordered the elements. (b) Use the masses and atomic numbers of Ar and K from your Element Cards, and what you know about how each reacts, to explain why the modern table is ordered the way it is. (c) Define a group and a period, and describe where Li, Na and K and where Ne and Ar fell when you arranged the cards by atomic number. (d) Look at the masses of Li, Na and K on your cards, and at Ca and Ar. Why was mass alone not enough to explain the patterns you saw?

Write your answer here:

Part 2: Electron Arrangement, Group and Period

(a) Write the electron arrangement of each of these atoms in both energy-level form and s and p notation: Li, Na, Cl, Ca. **(b)** State the rule that links the electron arrangement to the group number for groups 1 and 2 and for groups 13 to 18, and the rule that links it to the period number. Say what is special about He. **(c)** Use these rules to state the group and period of Na, Cl and Ne. **(d)** Use your answers to explain why Li, Na and K sit in the same column although their masses are very different, and why neighbours such as Na and Mg do not behave alike.

Write your answer here:

Part 3: Families, Stability and Reactivity

(a) Name the family of each: Li, Mg, Cl and Ar, and give the number of outer electrons of each family. (b) Describe what you saw when Li, Na and K were added to water, and state the reactivity trend down group 1. (c) Use the electron arrangements of Ne (2,8), Ar (2,8,8), Na and Cl to explain why Ne and Ar do not react while Na reacts vigorously and chlorine is used for water treatment in Nairobi and Mombasa.

Write your answer here:

Part 4: Predicting Ions, Valency and Formulae

(a) For each of Na, Mg, Al, O and Cl, write the ion formed with its charge and its electron arrangement in s and p notation, and state which noble gas arrangement it matches. (b) What is the oxidation number of Na, Mg, Al, Cl and O? Explain how it differs from valency. (c) Use the criss-cross method to derive the formulae of calcium chloride, aluminium oxide, and the fertilizer made from potassium and the nitrate radical, used on farms in Kericho and the Rift Valley. Use your Radical Cards from Lesson 8. (d) Why can outer electron count alone not predict the ion of iron or copper? Use the compounds from Lesson 7 as evidence.

Write your answer here:

Part 5: Putting It All Together and Predicting Strontium

Strontium (Sr, Z38) is an element you have never worked with. Its electron arrangement is 2,8,18,8,2. (a) Use this arrangement and the rules you have built to predict its group and period, and explain how you decided. (b) Predict the ion it forms and the formulae of its compounds with chlorine and oxygen. (c) Write a balanced equation with state symbols for sodium reacting with water, and explain why potassium and lithium follow the same pattern. (d) Write a final explanation of the driving question: how does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds they form?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Ordering and history of the periodic table	Correctly explains Mendeleev (mass, gaps) and Moseley (atomic number), and uses the Ar and K data to justify ordering by atomic number.	Describes both scientists correctly but gives limited use of the Ar and K evidence.	Mixes up the scientists or the basis of order, or gives no evidence.
Electron arrangement, group and period	Writes correct arrangements in both forms and applies the group and period rules accurately (including the He exception) to explain why Li, Na and K are alike and Na and Mg differ.	Writes mostly correct arrangements and states the rules, with small errors in applying them.	Arrangements or rules are incorrect or not linked to group and period.
Stability and reactivity reasoning	Uses full outer levels, losing and gaining electrons, and specific evidence (water reactions, Ne, Ar, Cl) in a clear claim-evidence-reasoning explanation.	Explains stability and reactivity with some evidence but reasoning is incomplete.	Describes observations without linking them to electron arrangement.
Predicting ions, valency and formulae	Correctly gives ions with their arrangements, oxidation numbers, formulae, a balanced equation with state symbols, and explains the limits for iron and copper and the prediction for strontium.	Mostly correct predictions with minor errors in charges, formulae or balancing.	Many incorrect ions, formulae or an unbalanced equation.

Scientific communication	Complete sentences, accurate vocabulary and Kenyan examples, and a coherent final explanation that answers the driving question.	Mostly clear with correct vocabulary but some gaps in the final explanation.	Unclear writing, vocabulary misused, final explanation missing key ideas.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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